Derivation of German Logistics Data

INTRODUCTION

This paper serves as the bibliographic reference for sources used to compile data on German logistics in the Ardennes Campaign Simulation Data Base (ACSDB). Because of a lack of primary and secondary source information on German Army Group B logistics in the Ardennes Campaign, the majority of the data on German logistics in the ACSDB was estimated. Therefore, this paper also explains all data estimation methodologies used to generate German logistics figures.

As explained in the ACSDB Unit Data Base narrative, Fields 43-48, and 54-55 of the Unit Data Base were used to record information on ammunition, other supply, and motor fuel or petroleum, oil, and lubricants (POL) amounts -- stored, received, and expended or consumed by ground forces. All ammunition data requirements for the ACSDB were generated for German forces. However, no data on "Other Supply," i.e., food, fodder, medical supplies, and other miscellaneous supply, was estimated for German units in the ACSDB. Likewise, data on "Transportation Capacity (dry)/Transportation Capacity (wet)" was estimated only for corps-level units. A later section of this paper discusses the dearth of information in primary and secondary sources on German transport capacity in the Ardennes Campaign, and the expediencies adopted by the Germans to move their supplies from dumps to frontline units, as well as recommendations for possible data substitutes which may be used in lieu of hard data on German transport capacity.

The paper is divided into three major sections. The first addresses German ammunition data generation and the conventions used for recording the data in the ACSDB. The second section provides information on German fuel data and its generation and the format for recording it in the ACSDB. The third addresses long-haul transport capacity for German units.

Full bibliographic information is provided either in this paper or in the Bibliography Data Base for all primary and secondary sources referenced in the paper.

To assist in the review of German logistics data recorded in the ACSDB, the following discussion of German supply doctrine and assumptions made for the ACSDB is provided. By German doctrine, the movement and allocation of supplies was the responsibility of the army. Supplies were transported from railheads by army transport to army dumps and then distributed to corps

distribution points by army and corps transport. (The corps augmented transport capacity of the army/divisions as the situation required.) Division transport then picked up supplies from distribution points and moved them to the divisional dump. In some cases supplies would be transported directly from army dumps to division dumps by army and/or divisional transport. In the Ardennes Campaign, this doctrine was modified. Apparently the army dumps established prior to the German offensive were the same as the railheads in the army group area. Due to the lack of transport capacity at all echelons, German divisions in the Ardennes Campaign drew supplies either directly from the these railheads, or, in some cases, from supply dumps east of the Rhine River. This meant that a divisions could on one day draw supplies from a railhead as little as 5-10 kilometers away, and on the next day, due to interdiction of rail communications, be forced to draw supplies from dumps over 100 kilometers away. An additional supply complication arose because the supply dumps were limited to existing railheads and the rail network at the start of the Ardennes Campaign, and supply distances increased as units advanced, further overloading an already overtaxed transport system. (Due to manpower shortages and Allied air supremacy, it proved impossible to extend the rail net forward as units advanced.)

GERMAN AMMUNITION DATA

Information on categorization of German ammunition types is found in the ACSDB T/O&E (Table of Organization and Equipment) Data Base narrative. This categorization scheme was utilized for German T/O&E and Unit Data Base logistics data in the ACSDB.

Sources Used for German Ammunition Data.

A variety of sources were used to derive German ammunition data for the ACSDB. It is recommended that some of them be reviewed by users of the ACSDB for better comprehension of the complex subject of German logistics, which unfortunately lacks the kind of complete and detailed documentation most suitable for the data compilation tasks of the ACSDB. Chief among these sources is a manuscript study prepared by Charles V. von Luettichau in September 1952 for the Office of the Chief of Military History (Charles V. von Luettichau, GERMAN RAIL COMMUNICATIONS IN THE ARDENNES OFFENSIVE 1944-1945, A Study of the Factors Which Affected the Strategic Concentration and the Conduct of Operations (Washington, D.C.: Office of the Chief of Military History, September 1952)). Information contained in this study was used extensively in the ACSDB estimation process for German logistics data (the study is hereafter referenced in this paper as "von Luettichau, GERMAN RAIL"). Two other sources which should be reviewed and were also used for data estimation are daily supply reports for the German XLVII Panzer and LXXXI Korps. The first are a series of daily XLVII Panzer Korps reports produced by the Korps Quartermaster for the period 14-31 December 1944 on US National Archives Microfilm T314,1133,626-644. The second document is the "LXXXI Korps Munitions Situation according to the Daily Report for the period 15-31 December 1944...Total Rounds Fired in Tons" ("Munitionslage nach der Tagesmeldung...Gesamtverschuss in to:") on US National Archives Microfilm T314,1594,1131-1330. Both Korps documents are the only primary source records found in research for the ACSDB, which contain information on daily ammunition expenditures by German units below army-level in the Ardennes Campaign.

In addition to information found in these three sources, a variety of other sources, primary and secondary, were researched and examined for German logistics data. Attachment 1 contains a list of these sources and a brief description of information contained in them. It is recommended that they also be reviewed if users of the ACSDB want fullest appreciation of all available primary and secondary source data on German logistics in the Ardennes Campaign.

Estimation Methodology for German Ammunition Data.

The data estimation methodology for German ammunition supply in the ACSDB was created 1) to reflect as completely as possible all information contained in sources on German ammunition supply, and 2) to permit a logical and mathematically sound approach for data generation and recording in computer format. Due to the unavailability of daily historical German ammunition statistics, it proved necessary to estimate virtually all ammunition data for the German armies, corps, divisions, and non-divisional units in the ACSDB. The fragmentary information found in the historical record was used to the extent possible to formulate the estimation methodologies for German ammunition data.

The first step in estimating German ammunition data was to determine the initial and total amounts of available ammunition for units, using all available information found in the historical record. Several of the Foreign Military Studies produced in the postwar period by German participants in the Ardennes Campaign discussed the supplies of ammunition stockpiled for the offensive. These sources focussed primarily on artillery ammunition supplies. (See Attachment 1 for a listing of sources.) Other primary sources (also listed in Attachment 1) had information on ammunition supplies stockpiled at army group and for the High Command of the West (OB West) in December 1944.

From these sources, ammunition levels were estimated for the non-divisional, divisional, corps, and army units of German Army Group B. The total amount of ammunition expended by Army Group B during the period 16 December 1944-16 January 1945 was estimated to equal one issue of supply for each division and non-divisional combat unit. This assumption applied to Tank/AT (Antitank) and Other ammunition for all 5th Panzer, 6th Panzer, 7th, and 15th Army units, and to Artillery and Special ammunition for 15th Army units. For Artillery ammunition of the 5th Panzer, 6th Panzer, and 7th Armies, the amount expended was assumed to equal one issue of supply, plus 3/8 of an issue per day expended on the first two days of the offensive for the 7th Army, and 1/2 of an issue per day on the first two days of the offensive for the 5th and 6th Panzer Armies.

Ammunition weights by unit for one issue of supply were obtained from the Tables of Organization and Equipment compiled for German units in the ACSDB. TM-E 30-451, pages VI-20 to VI-21 (source referenced in the Bibliography Data Base) contained information on German ammunition allowances in World War II. A discussion in the ACSDB T/O&E Data Base narrative compares the information found in TM-E 30-451 with data found in another source on German divisional ammunition levels (see RH3/v.135, a source referenced in the Bibliography Data Base). As explained in the ACSDB T/O&E narrative, the authorized ammunition supply weights in German T/O&Es were assumed to be close to the maximum

weights maintained by German units in the Ardennes Campaign.

Each division was assumed to start on 16 December, or its first day of participation in the ACSDB, with three-quarters of an authorized issue of supply for each ammunition type carried by the division type. For artillery units of the 5th and 6th Panzer Armies, and for the divisional artillery of divisions initially subordinated to these armies, the initial artillery ammunition weights included three-quarters of an issue plus one entire issue for the artillery preparation. For artillery units of the 7th Army, and for the divisional artillery of divisions initially subordinated to this army, the initial artillery ammunition weights included three-quarters of an issue plus three-quarters of an issue for the preparation. Note that for non-divisional units of all armies, data was estimated only for the unit's primary type of ammunition, i.e., Artillery with artillery units, Tank/AT with panzer (armored) and panzerjaeger (antitank) units, Special with nebelwerfer (rocket projector) units, etc. Also note that the primary ammunition type of combat engineer units was considered Other, and for certain FLAK (antiaircraft) units both Other and Tank/AT, dependent on the armament of the FLAK unit and its ammunition types as listed in the T/O&E Data Base.

The second step in estimating daily German ammunition statistics was to determine ammunition consumption for German units. For each division and non-divisional unit type, i.e., volks grenadier division, panzer division, FLAK brigade, heavy panzerjaeger battalion, etc., "pools" of ammunition to be expended during the period 16 December-16 January, by type (Tank/AT, Artillery, Other, and Special), were calculated by multiplying the authorized issues of supply of the unit type by the number of the unit types which participated in the Ardennes Campaign. These were the total amounts of ammunition assumed to have been expended by the units during the period 16 December-16 January. The ammunition "pools" were segregated by division and non-divisional unit type to simplify the accounting process used to estimate daily ammunition consumption.

The data found in the XLVII Panzer and LXXXI Korps documents referenced above was used as the foundation for estimation of German ammunition consumption. Based on the ammunition consumption figures found in these sources, the following daily average consumption figures were calculated:

XLVII PzK: 3,104.8 metric tons/15 days (17-31 Dec 44), or 206.99 tons per day.

LXXXI K: 994.8 metric tons/17 days (15-31 Dec 44), or 58.52 tons per day.

These two corps were essentially composed of the same numbers of divisions and non-divisional units (three-four

divisions, plus corps artillery) during the second half of December 1944. Therefore, the ratio of their daily ammunition averages were estimated to be the ratio of ammunition consumptions for an "active" day and an "inactive" day (206.99/58.52, or 3.5:1 after rounding). The XLVII Panzer Korps was one of the primary attacking corps of the German offensive, and the LXXXI Korps was a 15th Army corps which was engaged in static, positional warfare east of Aachen during the second half of December.

The total numbers of active and inactive days by division and non-divisional unit type were determined next. The daily activity levels were the same as used in the German battle casualty estimation methodology (see the German Divisional Battle Casualty Estimation Methodology narrative for a list of daily activity levels for each division). Activity levels of non-divisional units were determined in the cases of attachments to divisions, by the activity level of the parent division, and, in the cases of subordination to corps and army, assumed to be "inactive."

Using the ratio of 3.5:1 and the total number of active/inactive days by unit type, daily average ammunition consumption figures for an active or inactive day were estimated for the various division types. For example, there were four SS panzer divisions (1st, 2d, 9th, and 12th) in the Ardennes Campaign (the 10th SSPzD was not engaged in the combat operations, except for its artillery in the preparation fire), which together had an estimated 46,601.20 hundredweight of Other ammunition to be expended during the period 16 December-16 January (11,650.30 x 4). Together all four divisions had 77 active days, 34 inactive days, and 16 out of contact days during which the divisions were completely out of the front line. The total number of activity points was calculated as 303.5 ([77x3.5] + 34). Other ammunition expenditure for an SS panzer division on an inactive day was therefore estimated to be 153.55 hundredweight (46,601.20/303.5), and on an active day to be 537.43 hundredweight (153.55x3.5). On out of contact days ammunition consumed (expended) was assumed to be none (0).

For Artillery ammunition consumption of 5th and 6th Panzer and 7th Army units, the above-described procedure was estimated for the period 18 December-16 January only. For 16 and 17 December, 5th and 6th Panzer Army artillery consumption was estimated at one-half of an artillery issue of supply on each day, and three-eighths of an issue of supply per day for 7th Army units, these amounts based on descriptions of the artillery preparation fire found in sources listed in Attachment 1.

The third step in generating daily ammunition data for German units was to estimate on-hand ammunition amounts for divisions and non-divisional units. This estimation process was

based on descriptions in von Luettichau, GERMAN RAIL, pages 120, 126, 142 and 143. These descriptions indicate that, for all practical purposes, railroad lines of communications on which supplies were shipped to army and divisional dumps were severed during the period 29 December-4 January by damage to and destruction of railroad bridges and rail lines leading into the Ardennes region from the main supply dumps east of the Rhine River. Nevertheless, Army Group B apparently managed to maintain at least one day's supply of ammunition (see page 120, "...supplies on hand were sufficient for the next day's operations"). Based on the definition of issue of supply in TM-E 30-451, page VI-20 to VI-21 ("three issues sufficient to maintain an army for 8-10 days"), and the reduced allowances of ammunition levels maintained by German units in the Ardennes, the minimum amount of ammunition for a unit during the period 29 December-4 January was assumed to be one-quarter of an issue. For the period 16-28 December, it was assumed to be three-quarters of an issue. After 4 January, any depleted ammunition stocks of divisions and non-divisional units below one-half of an issue were restored to one-half an issue, and this level was assumed to be the minimum amount maintained by units from 4-16 January.

Ammunition amounts received by divisions and non-divisional units were always estimated to equal the amount needed to maintain a unit at its minimum supply level, as described in the preceding paragraph. Therefore, ammunition received by units in most cases was estimated to equal the previous days' ammunition consumed.

The fourth and final step in generating ammunition data for German units in the ACSDB was to aggregate data for divisions and non-divisional units and use these compiled figures for army ammunition data. German doctrine maintained that corps headquarters were employed primarily for tactical purposes, i.e., operational control of divisions and non-divisional combat units, and that the army was responsible for logistical support of divisions and non-divisional combat units. Ammunition consumed data for the four armies was estimated to equal the sum of the ammunition received by all subordinated divisions and non-divisional units, and expended by independent army units, such as artillery and armored battalions. Thus, the ammunition consumption figures for armies were intended to reflect ammunition consumed by independent army units, and ammunition supplied to units at lower echelon. This data was recorded with the "army troops," i.e., "5th PzArmy Troops," 6th PzArmy Troops," "7th Army Troops," and "15th Army Troops." The data on ammunition amounts on-hand and received by independent army units, i.e., non-divisional army assets, was also recorded with the "army troops."

Data representing ammunition in army supply dumps and received by army supply dumps was recorded with "army HQ" units,

i.e., those units identified as "5th PzArmy," "6th PzArmy," "7th Army," and "15th Army." Ammunition received was estimated to equal the amount necessary to maintain one issue of supply.

Each army's issue of supply was calculated based on the average daily consumption of the army (ammunition expended by independent army units and ammunition delivered to divisions and corps troops). The daily artillery and special ammunition consumption from the periods 19-28 December and 7-16 January were used in calculations for the 5th Panzer, 6th Panzer, and 7th Armies. For all other ammunition types in all four armies (including artillery ammunition with the 15th Army), the periods were 17-28 December and 7-16 January. Data from 16-18 December was not used for the 5th Panzer, 6th Panzer, and 7th Armies because of the high artillery ammunition consumption rates of the first two days of the Ardennes Campaign. Data for 29 December-6 January was not used for any of the armies because of reduced ammunition consumption due to the interruption of the flow of supplies during the period 28 December-6 January, when divisions were not drawing the majority of their ammunition from army stocks. The following figures were calculated:

AVERAGE
DAILY CONSUMPTION OF
AMMUNITION (100 lbs.)

Army	Artillery	Tank/AT	Special	Other
15th	3256.45	1181.83	0.00*	1343.84
5th	4867.13	4858.39	706.63	2522.35
6th	5611.38	4584.73	2318.85	2748.58
7th	2974.84	1517.56	885.31	1286.63

* No nebelwerfer (rocket projector) units with the 15th Army.

Miscellaneous Information on German Ammunition Data.

This section of the narrative contains miscellaneous information on the data compiled for German ammunition in the ACSDB.

- The 150th PzBde had no ammunition data estimated. It was assumed that majority of its supplies were drawn from 6th PzArmy divisions, principally the 1st SSPzD, alongside which it operated for most of its participation in the Ardennes Campaign.

- Ammunition data for the FBB and FGB (two armored brigades for which T/O&Es were not generated) was estimated based on the relative equipment strengths of the brigades and an army panzer division (PzD). The estimated percentages of panzer division ammunition data used for the brigades was as follows:

FBB: 1/3 Artillery; 1/2 Tank/AT; and 1 Other.
FGB: 1/2 Artillery; 1/2 Tank/AT; and 1 Other.

- Artillery ammunition data for immobile artillery units was calculated using the same conventions used for other artillery units, and the ammunition data estimated for the immobile artillery units was recorded with the "corps troops" units.

- Certain units lacked organic tank/AT and artillery weapons, including the 5th FJD (no artillery except 12cm heavy mortars -- HMTR); the 79th VGD and 85th ID (no self-propelled antitank guns); and the 326th and 560th VGDs (no self-propelled antitank guns until 26 and 25 December, respectively). Ammunition data for these units was modified accordingly, in the case of the units without self-propelled antitank guns by reducing the ammunition data by approximately one-third (the divisions retained towed antitank guns). For the 5th FJD, the ammunition data of the 406th VAK(-) was included with the division, as this volks artillery korps effectively replaced the division's organic artillery component during the Ardennes Campaign.

- All ammunition data for corps was recorded with "corps troops" units, i.e., "LVIII PzK Troops," "Korps Felber Troops," "LXXXI Korps Troops," etc. This data represents the ammunition of separate non-divisional combat units attached directly to corps. Ammunition data for divisional attachments was recorded with the divisional attachments ("Att"), i.e., "1st SSPzD Att," "167th VGD Att," "212th VGD Att," etc. This data represents the ammunition of non-divisional units attached directly to divisions. Divisional ammunition data (i.e., the ammunition data of organic divisional units only) was recorded with the division ("1st SSPzD," "276th VGD," "9th PzD," etc.). Data for armies and army troops was recorded in the format explained above. To determine the composition of corps and armies, and attachments to divisions, it is necessary to consult the "ATTACHMENTS & DETACHMENTS" section of the German Unit Data Base, in which is recorded the daily order of battle of German Army Group B during the Ardennes Campaign period.

GERMAN FUEL DATA

German fuel data in the ACSDB was estimated for consumption and supply of motor fuel only. Information in primary and secondary sources on German fuel supply in the Ardennes Campaign was characterized by a lack of detailed, daily data necessary to meet the requirements of the ACSDB. However, primary sources did contain aggregate statistics on fuel availability at the army group level, and secondary sources, mainly from the Foreign Military Studies series at the US National Archives, provided information on fuel levels available for certain attack divisions. The statistics and information were incorporated in the estimation process used to generate daily fuel data for the German units in the ACSDB.

Estimation Methodology for German Fuel Data.

The first step in estimating German fuel data was to estimate fuel consumption of German divisions. German divisional fuel consumption was based on fuel consumed in displacement (movement) of the division, and fuel consumed for administrative purposes, such as powering generators, stoves, etc., and for minor tactical movements.

Fuel consumption for displacement of divisions was based on the displacement of a division, as recorded in the Unit Location Data Base, and a fuel consumption factor for unit type derived either from the T/O&E fuel data of the unit, or from primary source records. T/O&E fuel data is the amount of fuel required to move all vehicles of a unit 100 kilometers over moderate terrain, multiplied by four or five for non-armored and armored units respectively (see the ACSDB T/O&E Data Base narrative for details). The T/O&E fuel amount is the amount planned for by German staff doctrine. The 100 kilometer distance was halved, for application in the fuel estimation methodology, to reflect information provided in multiple sources, and specifically in two sources (ETHINT 21 and A938) which indicated that the fuel supply necessary to move a unit 100 kilometers was adequate for only 50 kilometers of movement in the rugged terrain of the Ardennes. ETHINT 21, pg. 4, and A938, pg. 3 (both sources completely referenced in the Bibliography Data Base) both mention that one verbrauchssaetze ([fuel] requirement) was sufficient for only 50 kilometers of movement.

Fuel consumption for administrative purposes was estimated to be 0.0267 times the amount of fuel required to move a unit 50 kilometers. This fraction was based on information found in FM 101-10, Staff Officers' Field Manual: Organizational, Technical and Logistical Data, 21 December 1944, page 329, and the estimated fuel requirements for moving a German panzer division

50 kilometers in the Ardennes. FM 101-10, page 329, notes that 2,000 gallons of fuel was the normal daily requirement of a standard US armored division to power generators, cooking stoves, etc. Comparison of the amounts of fuel required to move a German panzer division and a US armored division the same distance showed that a US armored division used approximately twice the fuel to move the same distance. Therefore, the amount of daily administrative fuel requirements for a panzer division was estimated to be 1,000 gallons, or one-half of the US armored division requirements. This amount is 1,000/37,456, or 0.0267 of the 50-kilometer requirement of a panzer division. The value of 0.0267 was applied to the 50-kilometer requirement of all units to estimate the daily administrative fuel consumption of the units.

Attachment 2 is a list of fuel requirements for German divisions as used in the ACSDB.

Fuel consumption of divisions was calculated by determining the displacement of the division from data in the Unit Location Data Base, and using that value and the 50-kilometer requirement to arrive at fuel consumption for movement of the unit. To this value was added the administrative consumption to arrive at the total daily consumption of the unit. If a division's displacement was none (0.0 kilometers), it's fuel consumption was equal only to the administrative consumption.

The second step in estimating German fuel data was to calculate on-hand fuel supply of divisions. All divisions were assumed to start with one 50-kilometer fuel issue (the amount of fuel needed to move the unit 50 kilometers), unless data in primary or secondary sources indicated otherwise. This assumption was made based on the limited information found in sources on fuel supply of individual German divisions at the start of the Ardennes Campaign. These included:

Unit	Fuel (1,000 gals.)	Source
116th PzD	40.814	T314,1134,349
PzLehrD	63.360	T314,1134,276
26th VGD	7.075	T314,1134,296
12th SSPzD	60.593	B522, pgs. 12-13
1st SSPzD	56.041	B779, pgs. 14-15
FBB	65.712	B592, pg. 20
15th PzGD	45.988	T314,1134,520

(All sources described fully in the Bibliography Data Base.)

Analysis of these sources and the descriptions in other sources (see B151, pg. 147 and ETHINT 21, pg. 4, both sources fully referenced in the Bibliography Data Base), indicated that

divisional supplies of fuel probably ranged from one to two 50-kilometer fuel issues per first echelon division, with more than one issue in army dumps, but not with the front line units. German sources invariably mentioned a major shortage of fuel supply at division level, and so, based on these descriptions, and on the specified amounts of fuel for units in the above table, all divisions were assumed to have started with one 50-kilometer fuel issue.

Information on fuel levels maintained by divisions after the first day of the attack is minimal. Some sources described the capture of US fuel stocks, and this data, as well as any other information on daily on-hand fuel supply, is incorporated in the ACSDB. When no data was provided for a division in any source, the following methodology was used to estimate on-hand fuel supply and fuel receipts.

Divisions were assumed to have maintained at least three times the daily administrative fuel requirement (i.e., three days of supply for administrative purposes). Divisions were assumed not to have been resupplied with fuel until their stocks had been reduced to a level equal to three times the daily administrative fuel requirement. This assumption was made to reflect the chronic fuel shortages experienced by German units in the Ardennes Campaign. The minimum level of three times the administrative requirement was set on the assumption that a commander would not have allowed his unit's fuel supply to fall below this amount. Fuel received was then calculated as the amount necessary to maintain a unit's fuel level at three times the administrative requirement.

Historical fuel data was used in the ACSDB for the following units on the indicated dates. The source of the data is also provided. Note that the amounts for the 15th PzGD came from the same source shown in a previous table as having been used as the sources for on-hand fuel on 16 December. The amounts shown here were extrapolated to estimate the on-hand fuel of the 15th PzGD on 16 December, as shown in the previous table.

Unit	Date(s)	Fuel (1,000 gals.)	Source
116th PzD	21 Dec 44	30.000 captured	T314,1666,511
1st SSPzD	17 Dec 44	30.000 captured	ETHINT 11, pg. 2
2d PzD	22 Dec 44	14.482 on-hand	T314,1133,626
9th PzD	23 Dec 44	18.484 on-hand	T314,1133,626
PzLehrD	22 Dec 44	29.568 on-hand	SRH-049, pg. 133
PzLehrD	23 Dec 44	42.240 on-hand	SRH-049, pg. 133
15th PzGD	21 Dec 44	14.256 on-hand	T314,1134,520
15th PzGD	22 Dec 44	27.192 on-hand	T314,1134,520
15th PzGD	23 Dec 44	23.496 on-hand	T314,1134,520
15th PzGD	24 Dec 44	79.200 on-hand	T314,1134,520

(Sources in above table are fully referenced in the Bibliography Data Base.)

The third step in estimating German fuel data was to calculate daily fuel consumption, fuel received, and fuel amounts on-hand for divisional attachments and separate corps and army units. The methodology employed in this step was essentially the same as the one used for the divisions. Fuel data for divisional attachments was recorded with the records of units identified with "Att" in their title ("1st SSPzD Att," "167th VGD Att," etc.). Fuel data for corps and army separate units was recorded with the records of the corps and army headquarters units ("XLVII PzK," "5th PzArmy," "I SSPzK," etc.).

For divisional attachments, daily fuel consumption was based on the displacement of the units comprising the attachment and an administrative fuel consumption factor 0.0267 times the 50-kilometer fuel requirement. Divisional attachments are identified in the "ATTACHMENTS & DETACHMENTS" section of the Unit Data Base. The 50-kilometer fuel requirement was calculated in the same manner used for divisions. It was "tailored" to the daily order of battle of the attachments, i.e., the fuel requirement was driven by the composition of the divisional attachments. Displacement was determined by the displacement of the parent division of the attachments. No information was found in historical records on daily fuel supply for German non-divisional units. Therefore, because of this lack of information, and to simplify accounting procedures, on-hand fuel amounts and fuel receipts were assumed to equal the daily expenditures of the attachments. Implementation of this limitation on German fuel supply for non-divisional units was also done to reflect the chronic fuel shortages experienced by the German forces in the Ardennes Campaign.

For separate corps and army units, daily fuel consumption was estimated to equal the amount of fuel required to move, plus fuel required for administrative consumption. The displacement of corps and armies is recorded in the Unit Location Data Base.

The daily fuel consumption was calculated by adding the movement consumption of the corps or army units (as identified in the "ATTACHMENTS & DETACHMENTS" section of the Unit Data Base); the movement consumption of service support units of corps or armies; and administrative fuel consumption. The service support consumption factor was based on the amount of fuel required to move 123 PKW (personnel carriers) and 234 LKW (cargo carriers) the displacement distance of the corps or army.

Start on-hand fuel amounts for corps and armies were estimated to equal the first day's consumption of the unit plus a reserve of fuel. The reserve of fuel for all corps was based on the estimated daily administrative consumption of the I SSPzK (1,535 gallons) times three (1,535 x 3, or 4,605 gallons). All corps and armies were assumed to have maintained 4,605 gallons on-hand, and daily fuel receipts were estimated to equal the amount of fuel required to maintain this on-hand value. Note that the administrative consumption amount (1,535 gallons) was used for <u>all</u> corps at all times.

The fourth step in estimating German fuel data was to calculate army-level fuel supply figures. As German corps were by doctrine tasked primarily for tactical control of combat units, all German fuel dump stocks were assumed to have been maintained at army-level in the ACSDB.

T122, pgs. 410-414, noted that Army Group B had 12,000 cubic meters (CBM) of fuel stockpiled at the beginning of the Ardennes Campaign. This was estimated in the ACSDB to be the total amount of on-hand fuel with the four armies of Army Group B on 16 December 1944. From this total amount (3,168,000 gallons, or 12,000 CBM x 264 gallons per CBM), the amounts of fuel with all separate army and corps units, all divisions, and all divisional attachments were subtracted (746,665 gallons). The remainder (2,421,335 gallons) was considered the fuel in army dumps. It was distributed among the four armies using the percentages of on-hand fuel with the subordinated units of the armies. These percentages were:

- 15th Army: 8.47%
- 7th Army: 21.78%
- 6th PzArmy: 40.94%
- 5th PzArmy: 28.81%

These percentages were applied to the total army stocks (2,421,335), to which were added the amounts with the separate army units, to estimate the amount of fuel per army on 16 December.

- 15th Army:	218.409	(1,000	gals.)
- 7th Army:	533.462	"	"
- 6th PzArmy:	1,011.913	"	"
- 5th PzArmy:	704.816	"	"

Daily fuel consumption by army was calculated as the total of all receipts by subordinated separate army and corps units, divisions, and divisional attachments. To this total was added fuel estimated to have been consumed by army transport vehicles moving supplies (all fuel and ammunition and a portion of other supply -- rations, medical supplies, etc.) from army dumps to units .

The formula used for army transport vehicle fuel consumption was based on a payload capacity of 3 tons per truck, a fuel mileage of 42.24 gallons per 100 kilometer, and one-way supply distance of 100 kilometers (200 kilometers round trip).

The standard German wheeled cargo carrier had a 3-ton payload. RH10/363, pg. 10, gives the mileage of a German 3-ton truck [(m.LKW (o) (S. u. A. - Typ)] as 80 liters per 100 kilometers over moderate terrain. (This figure was doubled to 160 liters, or 42.24 gallons, per 100 kilometer to reflect increased fuel consumption in the rugged terrain of the Ardennes region.) The supply distance was an approximation and was based on the distance from the 16 December front line to the Rhine-Moselle River line.

To determine daily fuel consumption by army, the estimated daily amount of fuel required to move all fuel and ammunition received by units subordinated to an army, was added to the daily amount of fuel receipts by all subordinated units. To this sum was added fuel estimated to move one-fourth of the other supply (rations, medical supplies, etc.) to arrive at the total estimated daily consumption of an army.

The value of one-fourth of the other supply was determined by comparing the recorded fuel consumed during the period 16 December-3 January (20,000 CBM) in von Luettichau, GERMAN RAIL, pg. 138a, with the total amount estimated consumed in the ACSDB during the same period, the latter amount only for ammunition and fuel shipment and ammunition receipts by subordinated units. This comparison showed a difference of approximately 2,160,000 gallons of fuel. Assuming that one man required 24.99 pounds of other supply per day (see FM 101-10, Staff Officers' Field Manual: Organization, Technical and Logistical Data, 21 December 1944, pg. 308, and the ACSDB T/O&E Data Base narrative), the amount of fuel estimated as required to move the total other supply used from 16 December-3 January, based on the sum of the daily personnel strengths of the armies and the requirement per man of 24.99 pounds, was equal to approximately one-fourth of 2,160,000 gallons. The remainder of the other supply was assumed

to have been moved by horse-drawn transport.

Once the 16 December on-hand fuel supply of the armies had been determined, and the daily consumption by army also estimated, daily on-hand supply per army after 16 December was estimated by subtracting out the daily consumption from the 16 December on-hand supply, and repeating the process for each subsequent day. Fuel received was allocated to the armies for the period 16-29 December based on information found in von Luettichau, GERMAN RAIL, pgs. 133-34, which gives the dates of arrival and destinations of 39 fuel trains during the period. The destinations were used to determine which army received the fuel. The amount of fuel carried by each fuel train was based on von Luettichau, GERMAN RAIL, pg. 134. which described a total of 50 fuel trains moving 12,500 tons of fuel. For the period 1-16 January, fuel receipts by army were estimated to have arrived in numbers and at intervals sufficient to prevent the depletion of army stocks.

GERMAN LONG-HAUL TRANSPORT CAPACITY

Estimation of German long-haul transport capacity for the ACSDB was complicated by a lack of data on organization and capacity of German corps and army service support units during the Ardennes Campaign. Another complication was caused by the chaotic condition of supply transport for German Army Group B in December 1944 and January 1945. Numerous German sources mention the adverse effects of Allied air supremacy on the German lines of communication. Supply difficulties were compounded by a lack of motor fuel available to the combat and supply units, the variety of trucks in use with the German Army in late 1944 (Czech, Italian, French, Russian, German, etc.), the poor condition of motor vehicles and the lack of spare parts for the different truck models, shortages of trained, experienced drivers, and poor road conditions in the Ardennes region. German army-level transport shortages were so acute that combat divisions were forced to employ their organic transport for the retrieval of supplies from distant supply dumps, some located east of the Rhine River, a fact attested to in many of the post-war manuscripts prepared by German officers who participated in the Ardennes Campaign and on file at the US National Archives. (These and other sources are listed in Attachment 1 to this paper and in the Bibliography Data Base. They should be reviewed for details on the German transportation system in the Ardennes.) German transport difficulties were further evidenced by the fact that transport vehicles of some German combat formations were requisitioned and provided to first echelon formations to enhance their mobility.

With these considerations, estimation of German long-haul transport capacity started with the formulation of several possible estimation methodologies. These included:

1) Using the maximum tonnage of supplies moved by an army on a single day as the long-haul transport capacity of the army.

2) Using the average daily tonnage of supplies moved by an army as the long-haul transport capacity of the army.

3) Using the daily tonnage of supplies moved by an army, on a daily basis, as the long-haul transport capacity of the army.

4) Using the known transport column tonnage capacities of German armored or armored infantry divisions, or the I SS Panzer Korps, to estimate the long-haul transport capacity of armies. (This data is found in German Archives sources RH10/141, /148, /149, /178, /181, /309, /312, /313, /318,

/319, and /321, described in full bibliographic annotation in the Bibliography Data Base.)

Of the four alternative methodologies, (3) was considered the most applicable, because it appeared that, based on descriptions found in sources, German long-haul transport capacity fluctuated on a daily basis and was more a function of available transport vehicles than a standard organizational capacity.

The other three alternative methodologies were not considered applicable. Methodology (2) would have resulted in German supplies shipped exceeding the long-haul capacity on some days, while (1) would have inflated the capacity to a higher level than the Germans were probably able to maintain without interruption. No information was found on authorized long-haul transport capacity of army-level transport units. OKW, KTB, IV 2, pg. 983 (source fully described in the Bibliography Data Base), mentioned that Army Group B's total transport capacity for the Ardennes operation was supplemented by an additional 1,000 tons, but the distribution of this tonnage was not indicated, nor is it apparent whether the supplemental capacity in fact became available during the Ardennes Campaign (see von Luettichau, GERMAN RAIL, pg. 139). Using the capacities of the division transport columns to estimate data for army columns was deemed unacceptable, because 1) they were organized for different functions than the army units, and 2) there was no information relating divisional transport column capacities to that of their army counterparts. Although the I SS Panzer Korps document (RH10/309) contained information on transport column capacity for the corps, there was likewise no way to relate this capacity to that of its army counterpart.

Therfore, it was decided after deliberation to estimate long-haul transport capacity for corps units only. Using the daily tonnage of supplies used by an army was considered too artificial a construct for estimating German long-haul transport at the army level. There would be too much variation in the capacity and no possibility of distinguishing capacity provided by division transport from capacity of army transport columns.

Thus, for all corps assumed to have a transport component (see the German Non-Divisional Service Support Unit Personnel, Equipment, and Medical Data narrative for details), a dry transport capacity of 107 metric tons (converted to 117.914 short tons) was estimated, based on the column capacity of the I SS Panzer Korps on 1 December 1944. German Archives source RH10/309 (full reference in the Bibliography Data Base) is the source used for this capacity. The tonnage amount was not divided into wet and dry transport capacity, as there was no information on which to make such a division.

Attachment 2

Motor Fuel Requirements (in 1,000 gallons) for German Divisions

Unit	VS (50 km)	Admin. Use
1st SSPzD	46.701	1.247
2d SSPzD	44.937	1.200
9th SSPzD	44.864	1.198
10th SSPzD	44.930	1.200
12th SSPzD	46.610	1.244
PzLehrD*	42.240	1.128
116th PzD**	40.313	1.076
PzD (all others)	37.456	1.000
PzGD	25.726	0.687
26th VGD***	7.075	0.189
VGD (all others)	6.020	0.161
ID	7.952	0.213
27th & 28th SSPzGDs****	1.505	0.040
150th PzBde*****	5.562	0.148
FBB*****	32.856	0.877
FGB*****	34.649	0.925
3d FJD******	7.952	0.213
5th FJD******	6.020	0.161

* 50 km requirement from US National Archives microfilm T314,1134,276, 10 December 1944 Report of Quartermaster "Work Staff" Kollmann (special planning staff of XLVII PzK for Ardennes offensive).

** 50 km requirement from US National Archives microfilm T314,1134,349, 2 December 1944 Level of One Fuel Requirement and Fuel Stocks.

*** 50 km requirement from US National Archives microfilm T314,1134,296, 10 December 1944 Description of Fuel Situation from Quartermaster Report.

**** 50 kilometer requirement estimated to be one-fourth of VGD, based on numbers of vehicles in units.

***** 50 kilometer requirement estimated based on numbers of vehicles in units.

****** 50 kilometer requirements based on requirements of ID and VGD. Both fallschirmjaeger divisions (FJDs) were considerably below authorized strength in motor transport. (In B023, pg. 25, the 5th FJD was noted as lacking its supply columns entirely, which had to be replaced by 100 civilian horse-drawn vehicles)

All other values are calculated using the methodology discussed in the main body of this paper.

Compilation of German Order Of Battle

The task of compiling a daily German Order of Battle for the Ardennes Campaign Simulation Data Base (ACSDB) was complicated by the fact that information from primary and secondary sources was fragmentary and in some cases contradictory. This paper describes the nature of the German Order of Battle, problems encountered in its compilation, and the decisions made to create as accurate a final product as possible within the limitations of the existing records.

The sources referenced in the paper are either described in full bibliographic notation in Attachment 1 of this paper, or are listed in the Bibliography Data Base.

German Doctrine.

German Army doctrine allowed for a number of different types of subordination within the army's command structure. These included:

1) Attached for supply purposes only
2) Attached for tactical purposes
3) Subordinated for tactical purposes

Of these three the first is self-explanatory and is not considered an attachment for the purposes of the Order of Battle. The second form of subordination meant that the unit was for all intents and purposes a part of the parent unit. That is, supply was drawn from the parent unit, orders were given by the parent unit, and unit reports were passed via the parent unit. The third from of subordination is most closely analogous to the concept of OPCON (operational control). The subordinate was under operational control of a headquarter other than its parent unit but could continue to draw supplies and pass reports via its parent unit. By German doctrine, any unit under the command of another for a period less than eight weeks was considered to be subordinated. For the purposes of the ACSDB German Order of Battle, both the second and third forms are considered to be attached. An additional method of subordination employed by the Germans in the Ardennes and considered in the ACSDB was the formation of "ad hoc" headquarters to which divisions and other units were subordinated. For example, Korps Felber was formed around the headquarters staff of an infantry division, while Armeegruppe Von Luettwitz was formed by subordinating the XXXIX Panzer Korps to the XLVII Panzer Korps. (See the notes on Korps Decker below.)

Sources Description.

Primary sources used included the Heeresgruppe (Army Group) D Kriegstagebuch (War Diary) for December 1944. This unit, although not involved in the Ardennes, included in its war diary most of the daily situation reports disseminated by Heeresgruppe B during December 1944. These situation reports included fragmentary information on German headquarters locations, unit boundaries, and changes of assignment for various units. This information was utilized extensively for the ACSDB German Order of Battle. A second primary source was the Oberkommando der Wehrmacht (OKW) (High Command of the Armed Forces) situation maps. These provided information on assignments of many non-divisional units at the army level. Changes in assignment from one army to another, as identified in the OKW maps, were generally incorporated in the ACSDB Order of Battle, except in cases where changes directly contradicted other primary source material (see below). A third primary source used was the various Heeresgruppe B records dealing with preparations for the Ardennes Offensive. These included various Order of Battle charts and lists, some of them contradictory (see below). A fourth primary source used were extracts from the daily Allied intelligence summaries ("Sunset") derived from ULTRA intercepts. These contained information and data on the German Order of Battle, as well as other aspects of the German forces in the Ardennes. A final primary source used was the various corps and division reports generated by German units in the Ardennes. These included war diary excerpts, unit condition reports, and corps supply and transportation records. Units covered in these reports included: XLVII Panzer Korps, LIII Korps, LVIII Panzer Korps, LXXXI Korps, and various divisions.

Secondary sources used included German postwar manuscripts, written by German officers who participated in the Ardennes. The most valuable manuscripts were generally those compiled by the various army and corps artillery commanders. They proved to be the most consistent, complete, and accurate when compared with primary sources.

Other secondary sources included German postwar histories, of which that for the Panzer Lehr Division proved to be the most complete and accurate. The history of the II SS Panzer Korps was limited in its usefulness by the fact that it is restricted to the activities of the 2d and 9th SS Panzer Divisions.

Problems Encountered in Compiling the Order of Battle.

In addition to the lack of complete primary source information on a daily basis for the German Order of Battle, a major problem encountered in determining the Order of Battle was that in many cases two sources would be contradictory. For example, the OKW maps sometimes showed a change in subordination

several days after the change was noted in other primary sources. It is believed that this is a result of the apparent lack of timeliness in the maps. In cases such as this, the maps were considered to be incorrect. A major contradiction was found to be the assignment of the LVIII Panzer Korps in January 1945. This corps was active in the northwest corner of the Ardennes salient until 3 January. At that time the OKW maps show a change in location for the corps to the vicinity of Bastogne, then for the period from 5-10 January it disappears from the maps entirely. "Sunset" reports the relief of the corps by elements of the II SS Panzer Korps on 3 January and its reassignment to the vicinity of Bastogne. The postwar manuscript by the corps commander says that the corps moved to the vicinity of Bastogne on 11 January, whereas the corps artillery commander's manuscript says the move was made on 4 January. These contradictions were not resolved until a radio message dated 8 January was found on microfilm in the US National Archives (Microfilm #T314,1666,544), routed via the LVIII Panzer Korps signals battalion, with the sender listed as "Korps Decker." It was concluded that a temporary "Korpsgruppe" (corps group) under command of General Decker (commander of the XXXIX Panzer Korps which was being withdrawn on 5 January) was formed around the LVIII Panzer Korps artillery staff until the arrival of the remainder of the LVIII Panzer Korps headquarters on 11 January.

Other contradictions were found in the various planning papers of Heeresgruppe B for the offensive. A particular problem was the designation and assignment of some artillery units. In these cases the decision was made to use those documents with dates that were the closest to 16 December.

In cases where no resolution of a contradiction could be found, the original assignment of a unit was assumed to be correct. In cases where an effective date of change was missing it was assumed to have occurred on the day following the day on which a new assignment was first noted in a source.

Additional Notes on the German Order of Battle.

Several units have been eliminated from the Order of Battle, although they were originally assigned to the Ardennes Offensive. These include the 638th Army Artillery Battery (5th Panzer Army) and the 653d Heavy Antitank Battalion, both of which were apparently trapped in transit on railway cars by Allied bombing until they were transferred to Heeresgruppe G, as various rail transport notes in the Heeresgruppe D Kriegstagebuch for 21 December, 22 December, and particularly 29 December indicate.

Conventions Used in the German Order of Battle.

To reflect particular elements of the German tactical situation in the Ardennes in as accurate a manner as possible,

certain German units were tracked in a manner different from that used in the US segment of the ACSDB. The units concerned were all artillery, and the critical factor affecting the tactical situation was the mobility capability of these units.

The Germans used a number of different mobility classifications for their units, these included:

mot (motorisiert)	-	fully motorized
t-mot (teile motorisiert)	-	partly motorized
bew (bewegung)	-	mobile, horse drawn
t-bew (teile bewegung)	-	partly mobile, horse drawn
bo (bodenstandige)	-	non-mobile
Fst (festungs)	-	static fortress

Examination of available records showed that in the Ardennes Campaign even German units listed as motorized were stripped of vehicles to increase the supply transport capacity of corps or armies. Partly motorized or partly mobile units normally had the prime mover capacity to move one-third of the units' artillery pieces. To ensure that least a portion of the corps and army artillery would be capable of advancing to support the offensive, the German resorted to the expedient of concentrating the available prime mover capacity in a number of artillery batteries or battalions.

In order to reflect this fact, the following conventions in the German Order of Battle have been used.

1) All artillery units listed as attached to corps or army headquarters units (these units are designated as "6th PzArmy," "I SSPzK," etc.) are considered to be static; that is, they remained within the corps or army area as of 16 December 1944 and did not move forward with the corps or army combat units, if and when these units advanced.

2) All artillery units listed as attached to the Corps or Army "Troops" ("6th PzArmy Troops," "I SSPzK Troops," etc.) are considered to have some degree of mobility. For reference, the mobility classification of all artillery units is listed in the next section of this paper. That percent of the unit with a mobility capability should be considered to be able to move with the corps or army as it advances. That percent of the unit without a mobility capability should be treated as (1) above.

3) All artillery units, normally batteries and/or battalions, listed as attached to divisions should be considered to be <u>fully</u> mobile and should be treated as being capable of moving with the division to which it is attached. The sole exception to this is the artillery regiment of the 5th FJD, attached to the 352d VGD from 25 December to 16 January, which

was virtually immobile and only brought forward when the 352d VGD assumed a static defensive posture.

In the 16 December 1944 German Order of Battle, as recorded in the "ATTACHMENTS & DETACHMENTS" section of the Unit Data Base records, note that some organic divisional units, particularly with the volksgrenadier ("people's" infantry) divisions, are shown as being detached from the parent unit. This indicates that these components were not with the parent unit as of 16 December. This situation was particularly relevant to antitank components of a number of the volksgrenadier divisions. If the organic division unit's identification is followed by a "(REF)," it appears at a later date (between 17 December 1944 and 16 January 1945) in the data base. If the organic division unit is not followed by a "(REF)" on 16 December 1944, it does not show up in the Order of Battle on a later date, indicating that the parent unit did not have the unit within its organization during the Ardennes Offensive. For example, the 2d and 3d companies of the 79th Volks Grenadier Division's antitank battalion (2/ and 3/79th PjBN) are shown on 16 December as being detached from the 79th VGD. They do not appear on any date between 17 December 1944 and 16 January 1945 under "ATTACHMENTS & DETACHMENTS" and were therefore not with the division during the Ardennes Campaign.